Power Budget — Board B: Cap / Lid Sensing (Rev 3)

Team Number: 102
Project Name: Automatic Pill Dispenser
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Version: 3 (2026-10-27)

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	PIC18F57Q43 Curiosity Nano	DM164150 / PIC18F57Q43	+1.8 - 5.5V (VDD)	1	350	350	mA
	Quad op-amp	MCP6004-I/P	+1.8 - 6.0V	1	0.68	0.68	mA
	Dual voltage comparator	LM393P	+2 - 36V	1	2.5	2.5	mA
	Photoresistor divider	CdS LDR + 3.9k	+3.3V	1	0.83	0.83	mA
	Status + heartbeat LEDs	T-1 3/4 LEDs	+3.3V	2	4	8	mA
	Pull-ups + bias dividers	10k-470k 1/8W	+3.3V	1	1.8	1.8	mA

+12V Power Rail (not used	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	No +12V loads on Board B						

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	Curiosity Nano (VBUS via JP1)	DM164150 (MIC5353)	+5V	1	500	500	mA
	Power-good LED D2 + R1 (1k)	Green LED + 1k	+5V	1	3	3	mA
					Subtotal	503	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	628.75	mA
Regulator	+5V linear regulator	STMicroelectronics L7805CV	+7 - 35V in	1	1500	1500	mA

					Total Remaining Current Available on +5V Rail	871.25	mA

+3.3V Power Rail	**Component Name**	**Part Number**	**Supply Voltage Range**	**Qty.**	**Absolute Maximum Current (mA)**	**Total Current (mA)**	**Unit**
	PIC18F57Q43 (abs. max into VDD)	PIC18F57Q43	+1.8 - 5.5V	1	350	350	mA
	Quad op-amp	MCP6004-I/P	+1.8 - 6.0V	1	0.68	0.68	mA
	Dual comparator	LM393P	+2 - 36V	1	2.5	2.5	mA
	Photoresistor divider (worst case, bright light)	LDR + 3.9k	+3.3V	1	0.83	0.83	mA
	Status + heartbeat LEDs	D3, D4 (330 Ω)	+3.3V	2	4	8	mA
	Pull-ups + bias dividers	10k-470k 1/8W	+3.3V	1	1.8	1.8	mA
					Subtotal	363.81	mA
					Safety Margin	25%	
					Total Current Required on +3.3V Rail	454.7625	mA
Regulator	+3.3V LDO on Curiosity Nano	Microchip MIC5353	+2.6 - 6V in	1	500	500	mA
					Total Remaining Current Available on 3.3V Rail	45.2375	mA

External Power Source 1	**Component Name**	**Part Number**	**Voltage**	**Output Voltage**	**Maximum**	**Current**	**Unit**
Power Source 1 Selection	Plug-in wall adapter, regulated	Adafruit #63 (9V 1000mA)	100-240VAC	+9V	1000	1000	mA
Power Rails Connected to External Power Source 1	+5V regulator (draws +5V rail load)	L7805CV	+7 - 35V in	1	628.75	628.75	mA
					Total Remaining Current Available on External Power Source 1	371.25	mA

External Power Source 2: not used (Board B runs only from the 9V wall adapter)							

Notes:

Absolute maximum currents: PIC18F57Q43 max current into VDD = 350 mA (datasheet 47.1); MIC5353 LDO = 500 mA; L7805CV = 1.5 A; MCP6004 IQ max 170 µA/amp; LM393 ICC max 2.5 mA; LEDs (3.3 - 2.0)/330 ≈ 4 mA each.

Expected (typical) load is far lower: roughly 15-30 mA total, so the L7805 dissipates about (9 - 5) x 0.03 ≈ 0.12 W. At the worst-case 629 mA it would dissipate ≈ 2.5 W and need a heatsink.